A Semi-Fragile Watermark Scheme For Image Authentication

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Abstract

In this paper, a novel semi-fragile watermarking scheme for image authentication is proposed, which addresses the issue of protecting images from illegal manipulations and modifications. The scheme extracts a signature (watermark) from the original image and inserts this signature back into the image, avoiding additional signature files. The error correction coding (ECC) is used to encode the signatures that are extracted from the image. To increase the security of this scheme, user's private key is employed for encryption and decryption of the watermark during watermark extraction and insertion procedures. Experimental result shows that if there is no change in the obtained image, the watermark will be correctly extracted, thus will pass through the authentication system. This scheme is tolerant of lossy compression such as JPEG, but malicious changes of the image will result in the breach of the watermark detection. In addition, this scheme can detect the exact locations, which are illegal modified blocks.

Keyword: Image Authentication DWT BCH Private Key Semi-fragile Watermark

1. Introduction

The Internet is an excellent distribution system for the digital media because of its inexpensiveness and efficiency. However, the rapid distribution growth of digital media on the Internet has led to an urgent demand for the copyright and integrity protection because digital media can be easily replicated and modified. The existing methods for the intellectual property right protection are to embed robust digital watermarks into the multimedia data [1]. Many attempts have been proposed in literatures, but due to the wide variance of malicious attacks, the robustness of the watermark remains an unresolved issue.

Different from robust watermarking, the main purpose of the semi-fragile watermark is to identify illegal tamper of image rather than to verify its ownership, which is more important for medical image and photos used as court evidence. The previous techniques, such as [2], focused on detecting whether a whole image was tampered or not. But the algorithm in [2] did not clearly specify how and where the image was tampered. Friedman proposed a trusted digital camera, which embeds a digital signature for each captured image, which can be applied to verify the integrity of the image, as well as identify the specific camera that produced the image [3]. Jaejin Lee [4] adopts an error control coding (ECC) technique to generate the watermark, which in turn makes it possible to correct and detect the changes from the extracted watermark. But in this paper, ECC is used to correct the errors caused by noise and acceptable manipulation which may lead to change of the signature in the image. Another group of works change the least significant bit (LSB) of selected pixels to track the differences between randomly selected pairs of the points in [5]. Chen and Wang extract the content-dependent robust bits as the signature from spatial domain and insert this signature into the wavelet coefficients in [6]. The same as in [7], to survive after conventional compression JPEG compression, in an image authentication system, Lin and Chang proposed a method to preserve the invariance between the DCT coefficients before and after quantization to form a digital signature.

In this paper, we propose a new semi-fragile watermark scheme that can detect malicious modifications of the watermarked image and locate these tampered blocks. And this location is very useful, because the knowledge of where an image has been altered may be used to infer: I) possible motive of the tamper; II) possible candidate adversaries; and III) whether the alteration is legitimate [8]. In order to increase the security of this watermark system, a user's private key is used in this paper.

The structure of this paper is organized as follows. In the Section 2, we describe the proposed semi-fragile watermark scheme in detail. The Section 3 demonstrates some experimental results of scheme. Conclusions are given in the Section 4.

2. THE PROPOSED DIGITAL AUTHENTICATION SCHEME

We use a gray-scale image of size N*N pixels as the original image. The spatial domain method is used

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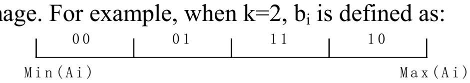
to extract the signature and the frequency domain method is used to insert the watermark. The wavelet transform is adopted in this paper due to its excellent multi-scale and precise localization properties.

2.1 Generate Digital Signature

First, the original image is divided into the non-overlapping blocks (16*16). Assuming there are n blocks after division and k bits signature is extracted from each block. Vector b_i (1*k) is defined as signature of block i. Thus the length of signature is k*n bits. We calculate the average gray value of each block A_i (i=1,2…n). The quantization step Δ is defined as:

$$\Delta = \frac{\max(A_i) - \min(A_i)}{2^k}$$

Gray Code is selected to encode the signature of the image. For example, when k=2, b_i is defined as:



When the length k is longer, the quantization step Δ is smaller, and the authentication system is more fragile. In this paper, k=4. Signature is the vector B=[b_1, b_2 ……b_n]. In [6], the quantization Δ is given by the sender. The receiver must know some information about the quantization to extract the watermark, which makes the security of the watermark scheme not high. But in this paper, the quantization step Δ is proposed through calculating the image's gray value, the receiver could detect the watermark without knowing anything about Δ.

After watermark being inserted back into the image, the average value of block A_i may be changed a little. For example, before insertion, A_i is slightly less than $\min + \Delta$, so b_i=00, but after insertion, A_i may be changed slightly more than $\min + \Delta$, b_i=01. Although the image hasn't been illegally manipulated, it couldn't pass through the image authentication system. On another hand, some acceptable manipulations such as JPEG lead to change the pixels of image. The extracted signature is not the same as the original signature. The obtained image also could not pass through the image authentication system. Therefore the signature is encoded by error correction coding (ECC). In [9], when the quantization Q must be less than $\Delta/2$, the obtained image was considered zero-error image authentication. In this paper, we select BCH (7,4,1) (Therefore the length of watermark in a block is 7) where one bit error could be corrected, and there is only one bit difference between the adjacent numbers in Gray code. Thus even the change of gray value of some block is no more than $3\Delta/2$, the ECC could help us to rectify the extract signature from the watermarked image. And we will show that these signature bits can be re-extracted from the degraded image, although the watermark insertion degrades the image a little in the experiment.

2.2 Insert Digital Watermark

There are many watermark insertion and detection algorithms by the wavelet transform. In most of these algorithms, watermark is inserted by modulating the wavelet coefficient. $W_{m,n}$ (x, y) is the selected wavelet coefficients at the scale m, the orientation n, and the position (x, y). $W_{L,L}$ is the main coefficients, and $W_{L,H}$ $W_{H,L}$ $W_{H,H}$ are the details of the image. So the watermark is inserted into $W_{L,L}$. Watermark is neither robust to be insensitive to malicious modification, nor fragile to be easily corrupted by reasonable compressions. Before signature is inserted, it is permutated by a user-key. The private key encrypts the output M. The W (x, y) is the selected coefficient, and the M (i) is the inserted watermark. Quantization step Q is selected by the scheme. The algorithm is described in the following structured codes:

```
For i=1:k*n
If M (i)=0 and mod (W (x, y), Q)=even
   W (x, y)=Q*round (W (x, y)/Q);
Else if M (i)=0 and mod (W (x, y), Q)=odd
   W (x, y)=Q*(round (W (x, y)/Q)+1);
If M (i)=1 and mod (W (x, y), Q)=odd
   W (x, y)=Q*round (W (x, y)/Q);
Else if M (i)=1and mod (W (x, y), Q)=even
   W (x, y)=Q*(round (W (x, y)/Q)+1);
End;
```

The watermark insertion scheme is shown in Fig.1

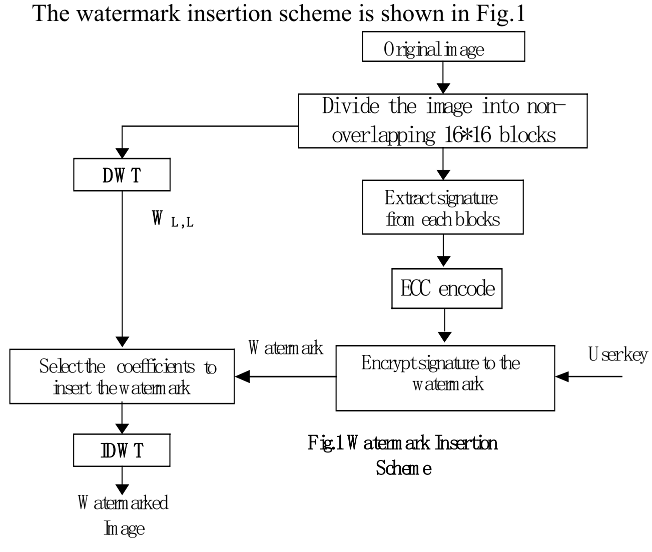


Fig.1 Watermark Insertion Scheme

Q is defined to be uniform in all coefficients. Assume the just noticeable change on a DWT coefficient is Q/2 and assume the largest applicable

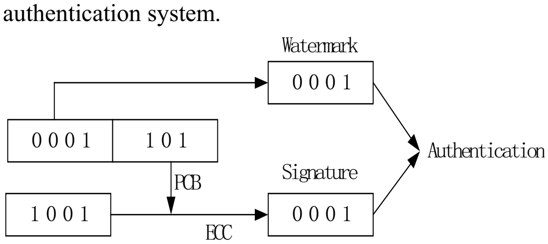
JPEG or other lossy attacks quantization step to this value is Q_m. By Shannon's well-known channel capacity bound, the capacity of this channel will be:

$$C(Q,Q_m)=\log_2\left(\frac{Q}{Q_m}+1\right)$$

For instance, the source coefficient value is I, and then its closest non-adjacency states of I are $I+Q_m$ and $I-Q_m$. To find out the JPEG capacity, we assume that all states within the just-noticeable range of $[I-Q_m, I+Q_m]$ the attacked image could pass the authentication system. Therefore, there are Q_m candidate JPEG or other lossy attacks states in this range. Since we have shown that the non-adjacent states have to separate from each other by Q_m, $Q_m<Q/2$. We will discuss how to select Q in the Section 3.

2.3 Verification Watermark Scheme

Obtaining the watermarked image, the receiver could extract the watermark from the image. First, we divide the obtained image into non-overlapping 16*16 blocks, and select the coefficients to extract encrypted codes after doing DWT in each block. Second, we use the private key to decrypt the signature from codes. This signature has two parts, the one is watermark, the other is parity-check-bit (PCB). By using this PCB to correct the signature, which is extracted from pixels of image, some error bits which is leaded by JPEG or other lossy attack could be corrected. For instance, the extracted code from DWT coefficients is 0001101, so watermark is 0001, and PCB is 101. But the extracted signature from pixels is 1001 and there is one bit error, which is not the same as 0001. So BCH is 1001101, and one error bit could be corrected by ECC. The final signature code is 0001, which is the same as watermark. But if the extracted signature code is 1101, which has 2 error bits, and BCH (7,4) could not correct 2 error bits, the obtained image could not pass authentication system.



Thus comparing the signature from the pixels of image and watermark from DWT coefficients, if the image is modified, the mismatched bits are detected by this algorithm. Otherwise, the receiver considers the image is integrity. Because the signature is extracted from the image block by block, if the receiver find some bits of watermark is wrong, these blocks are maliciously modified. The watermark authentication method is shown in Fig.2.

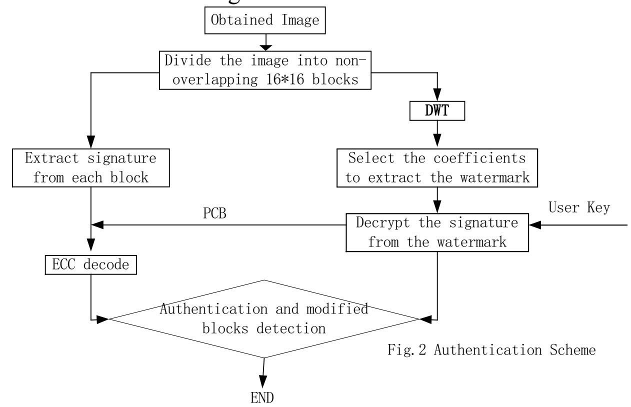


Fig. 2 Authentication Scheme

3. EXPERIMENT RESULTS

In this section, some experiment results are given to prove the efficiency of the proposed watermark scheme. We use the gray-scale three images--Lena, Couple and Jet (size 256*256, gray scale 0-255) as the original images. Comparing the original image and watermarked image, the change of image quantity after watermark insertion is invisible. The peak-to-peak signal-to-noise ratio (PSNR) is computed to evaluate the watermarked image quality. PSNR is defined as:

$$PSNR = 10\log_{10}\frac{255^2}{\delta_q^2}[dB]$$

Where the δ_q^2 is the mean square of the difference between the original image and the watermarked one. The PSNR of Fig 4 [b] is 47.12. Fig 3 shows the relation between the quantization step Q and the PSNR. X-axis is Q and Y-axis is PSNR.

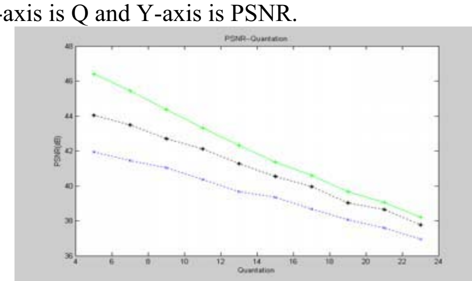


Fig.3 Quantization-PSNR

Furthermore, this proposed approach could be used for both fragile and semi-fragile image authentication by setting the value of the parameter Q to control the fragility of the system. The experiment result shows if Q=1,when the watermarked image is attacked by the Gauss noise, the watermark is fragile enough to detect this lossy attack.

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Fig 4(a) and (b) show the image before inserting watermark and after inserting watermark.

Fig.4 (a) original image   Fig.4 (b) watermarked image

In the experiment, some small blocks of the watermarked image are modified, the receiver could detect it by this scheme. Fig. 4[c] and Fig.4 [d] show the result of detections. The difference of signature and watermark indicate the modified block show in the Fig.4 [d], which are marked by white blocks.

Fig.4 (c) modified image   Fig.4 (d) located image

Fig.5 shows how JPEG compression influences algorithm. Whether the obtained image can be authenticated is decided by the quantization step Q. When Q=5, 87% JPEG compression image could pass through the authentication. When Q=20, even 45% JPEG compression image can pass, however, such compression leads to obvious image distortion. In Fig.5, X-axis is Q, Y-axis is acceptable JPEG compression quality factor.

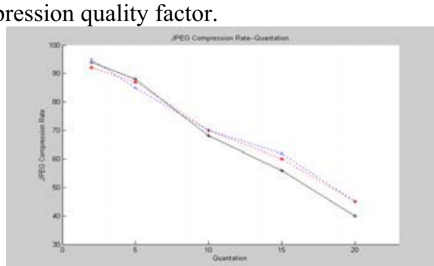

Fig.5 JPEG compression Experiment

4. CONCLUSIONS

In this paper, we present a new semi-fragile watermark scheme that accepts conventional image compression and rejects unacceptable manipulations such as crop-and-replace process. The embedded information is extracted from the original image. So when the image is obtained, even the receiver doesn't know anything about watermark, whether the image is illegally tampered is detected and the modified blocks are located by this scheme. The experiment results verified the effectiveness of the proposed scheme.

But in order to increase the security of the authentication system, we calculate the average value of the pixels of the each block. So if the block of minimum or maximum is modified, the quantization step Δ will be changed either. Although the modified image could not pass through the authentication system, the modified block could not be exactly located by this system. Our future directions include: more signature extraction method and possible ways to recover the illegally modified image without the original image.

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